

Experiment 16: Edge Detection using Canny Method

AIM

To detect the edges of an image using the Canny edge detection method.

PROCEDURE

1. Read the image.
2. Convert it to grayscale.
3. Apply the Canny edge detection.
4. Display the output image.

PROGRAM

```
import cv2

img = cv2.imread(r"C:\Users\chait\OneDrive\Documents\Desktop\cv lab image.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
edges = cv2.Canny(gray, 100, 200)
cv2.imshow("Original Image", img)
cv2.imshow("Canny Edge", edges)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

OUTPUT



Explanation

We converted the image to grayscale and applied the Canny edge detection algorithm. It detects the boundaries of objects by identifying sudden intensity changes.